

Machine-Learning Surrogate Optimization of a VPSA CO₂ Capture Process

SEEN5320 Final Project

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Abstract

This project builds a machine-learning workflow for a vacuum pressure swing adsorption (VPSA) CO₂ capture process using raw gProms simulation samples. The source file `data/raw_data.mat` contains 1708 successful simulations from an original 2000-point Latin hypercube design. Six operating variables are mapped to four scalar performance indicators: CO₂ recovery, CO₂ purity, productivity, and specific energy. Eight surrogate model families are tuned and compared on both global test accuracy and Pareto-focused accuracy. The best model, an RBF support vector regression surrogate, is then coupled with an in-house NSGA-II optimizer to generate surrogate Pareto fronts for purity-recovery and productivity-energy optimization.

1 Background and Objective

VPSA is attractive for post-combustion CO₂ capture because pressure-driven regeneration can be operated cyclically with relatively simple equipment. The reference study by Wang et al. [1] developed a 3-bed 7-step VPSA process for dry flue gas separation and used artificial neural-network surrogates to accelerate multi-objective optimization. The same project logic is followed here: detailed gProms simulations are treated as expensive samples, machine-learning models learn the input-output relationship, and a fast surrogate is used to search for improved operating conditions.

The project objective is to create a reproducible data-to-optimization pipeline:

1. convert the MATLAB simulation file into a clean CSV table;
2. train and tune several regression surrogates for the four VPSA KPIs;

3. evaluate accuracy globally and near empirical Pareto regions;
4. optimize the best surrogate with NSGA-II;
5. summarize the workflow and results in this report and the accompanying slides.

2 Literature Basis

Wang et al. [1] solved two VPSA multi-objective problems: maximizing purity and recovery, and maximizing productivity while minimizing energy under purity and recovery constraints. Their detailed process used silica gel adsorbent, a 15/85 mol% CO₂/N₂ feed, and six major decision variables: feed flowrate, replacement flowrate, vacuum flowrate, replacement time, equalization-depressurization time, and pressurization time. Their work motivates the same surrogate-assisted structure used here.

Placeholder for literature figure: 3-bed 7-step VPSA process schematic from Wang et al.

Figure 1: Literature process schematic placeholder.

Placeholder for literature figure: ANN architecture and surrogate-assisted optimization strategy.

Figure 2: Literature ANN and optimization workflow placeholder.

3 Data Conversion and Exploration

The raw MATLAB file contains two variables: `invar` with shape 1708×6 and `result_matrix` with shape 1708×284 . Only the first four columns of `result_matrix` are used as scalar KPIs. The conversion script writes `data/psa_scalar_samples.csv`, `data/psa_input_bounds.csv`, and `data/psa_scalar_summary.csv`. The final sample table has 1708 rows and 12 columns.

Table 1: Decision variables and sampling bounds.

Symbol	Meaning	Lower	Upper
FAD	Feed flowrate ($\text{m}^3 \text{h}^{-1}$)	0.50	1.00
FRP	Replacement flowrate ($\text{m}^3 \text{h}^{-1}$)	0.35	0.85
FVU	Vacuum flowrate ($\text{m}^3 \text{h}^{-1}$)	1.00	2.00
TRP	Replacement time (s)	35.00	55.00
TED	Equalization-depressurization time (s)	5.00	15.00
TPR	Pressurization time (s)	15.00	25.00

LHS Sampling Coverage for Successful gPROMS Simulations

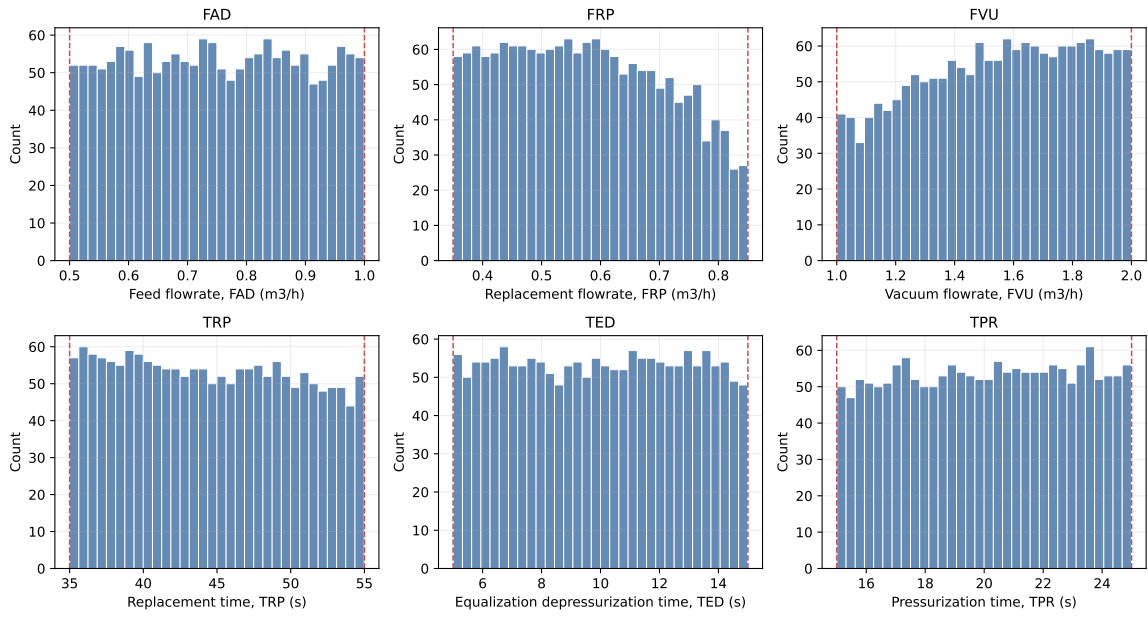


Figure 3: Sampling coverage of the six decision variables.

Distribution of Four Scalar PSA Performance Indicators

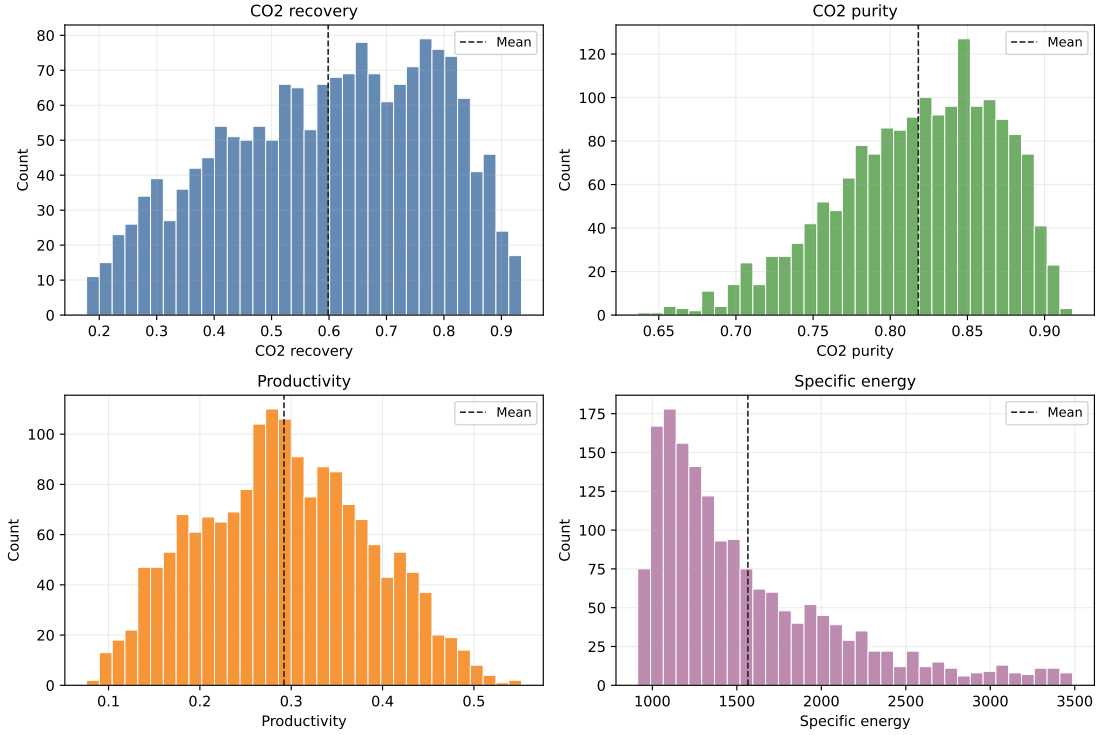


Figure 4: Distribution of recovery, purity, productivity, and specific energy in the successful simulation dataset.

The simulation data show the expected PSA tradeoffs. Recovery spans 0.178–0.935, purity spans 0.636–0.918, productivity spans 0.076–0.552 mol h⁻¹ kg⁻¹, and specific energy spans 913–3488 kJ kgCO₂⁻¹. The empirical fronts in Figure 3 are used to define near-Pareto subsets for model evaluation.

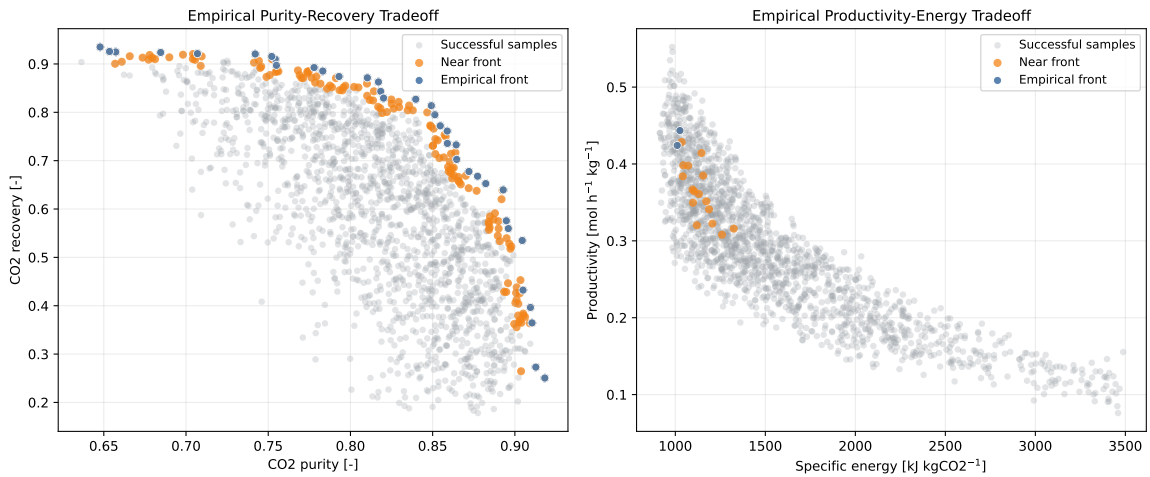


Figure 5: Empirical tradeoffs and near-front samples used for Pareto-focused validation.

4 Surrogate Modeling

The training script compares eight model families: ridge regression, polynomial ridge regression, RBF kernel ridge regression, RBF SVR, random forest, extra trees, histogram gradient boosting, and multilayer perceptron regression. The data are split into 80% training and 20% testing with random seed 42. Hyperparameters are selected by 3-fold cross-validation on the training set. Energy is modeled in log space to reduce scale imbalance, then transformed back to kJ kgCO_2^{-1} for reporting.

Models are ranked by

$$S = 0.6E_{\text{Pareto}} + 0.4E_{\text{global}},$$

where E is the mean normalized RMSE across recovery, purity, productivity, and log energy. This score emphasizes accuracy near the regions that matter most for optimization.

Table 2: Tuned surrogate comparison. Lower score is better.

Model	Global NRMSE	Pareto NRMSE	Score	Mean test R^2
RBF SVR	0.00276	0.00297	0.00288	0.99978
RBF kernel ridge	0.00255	0.00331	0.00301	0.99977
Polynomial ridge	0.00223	0.00510	0.00395	0.99983
Neural network	0.01160	0.01579	0.01411	0.99725
Gradient boosting	0.02497	0.02575	0.02544	0.98752

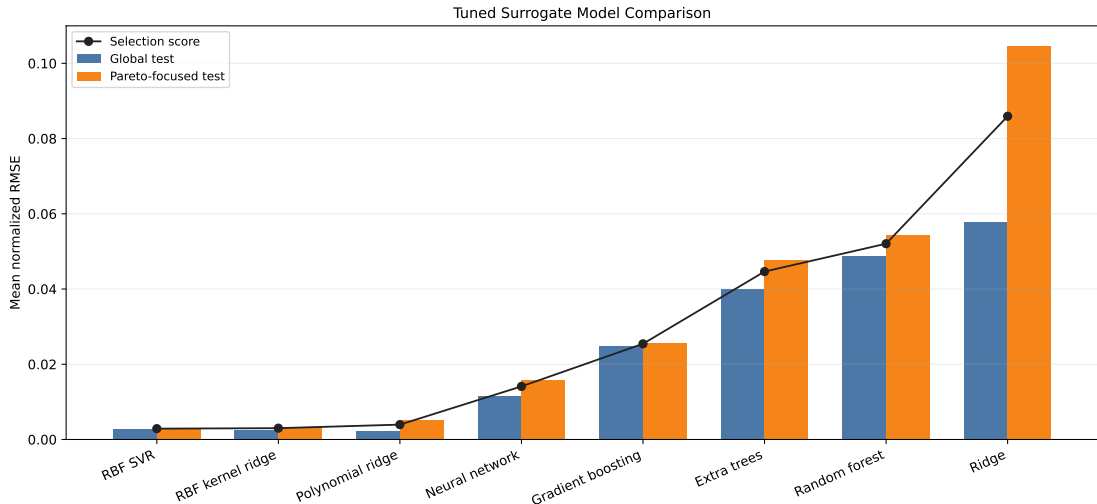


Figure 6: Global and Pareto-focused normalized RMSE for the tuned surrogate candidates.

The selected surrogate is the RBF SVR with $C = 100$, $\epsilon = 0.001$, and $\gamma = 0.1$. Its held-out test errors are small for all four KPIs: recovery RMSE is 0.00122, purity RMSE is 0.00035, productivity RMSE is $0.00064 \text{ mol h}^{-1} \text{ kg}^{-1}$, and energy RMSE is $22.74 \text{ kJ kgCO}_2^{-1}$. The predicted-versus-simulated plots confirm that this model is accurate enough for surrogate screening.

Best Tuned Surrogate: Held-Out Predictions

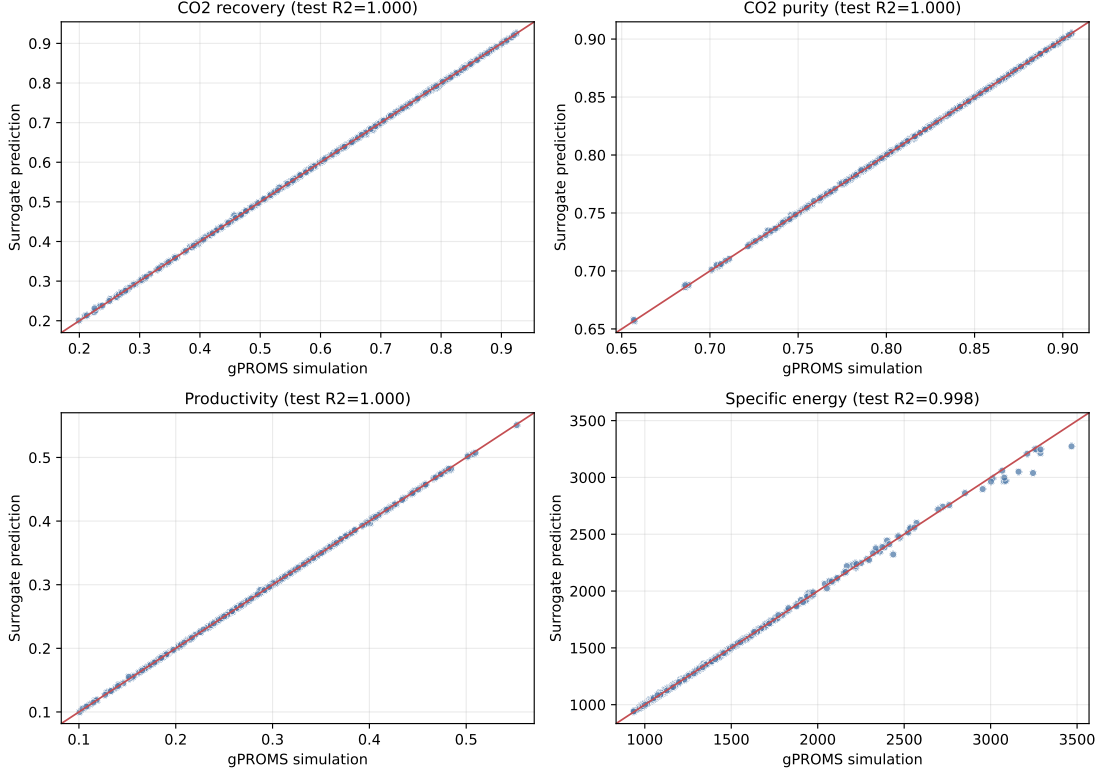


Figure 7: Best surrogate predictions on the held-out test set.

5 Surrogate-Assisted Optimization

The selected RBF SVR is optimized with an in-house NSGA-II implementation. Candidate solutions are represented as normalized decision variables and decoded to the physical bounds in Table 1. The optimizer uses simulated binary crossover, polynomial mutation, nondominated sorting, and crowding distance selection.

Two problems are solved:

$$\max (\text{purity}, \text{recovery})$$

and

$$\max \text{ productivity, } \min \text{ energy, } \text{ subject to } \text{purity} \geq 0.70, \text{ recovery} \geq 0.90.$$

Both runs use population size 160 and 250 generations.

Table 3: Surrogate optimization front summaries.

Problem	Purity range	Recovery range	Productivity range	Energy range
Purity-recovery	0.719–0.918	0.349–0.935	0.170–0.372	1127–2897
Productivity-energy	0.745–0.757	0.900–0.900	0.485–0.504	961–1003

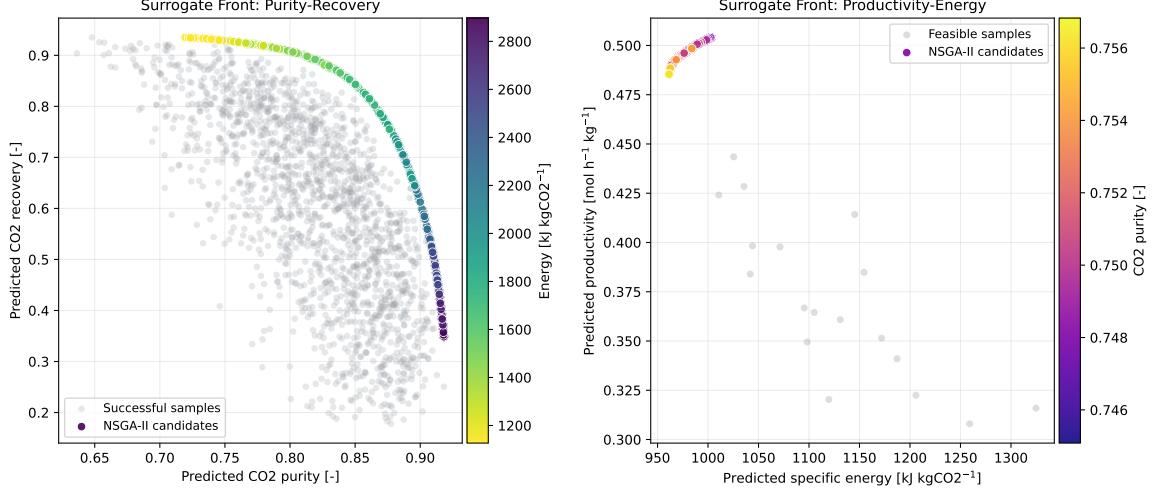


Figure 8: NSGA-II surrogate fronts for the two optimization problems.

6 Discussion

The purity-recovery front shows a strong compromise. The maximum-purity candidate reaches the observed upper purity bound of 0.918, but recovery falls to 0.349 and energy rises to about 2897 kJ kgCO₂⁻¹. The maximum-recovery candidate reaches 0.935 recovery with purity of 0.719, productivity of 0.358 mol h⁻¹ kg⁻¹, and energy of 1139 kJ kgCO₂⁻¹. This behavior matches the physical intuition from cyclic adsorption: pushing the bed toward higher enrichment can reduce capture recovery, while aggressive recovery can dilute product purity.

The constrained productivity-energy problem gives a narrower front. All nondominated candidates satisfy the purity and recovery constraints, and the optimizer concentrates near recovery of 0.900. The minimum-energy point predicts 0.756 purity, 0.900 recovery, 0.485 mol h⁻¹ kg⁻¹ productivity, and 961 kJ kgCO₂⁻¹ energy. The maximum-productivity point predicts 0.748 purity, 0.900 recovery, 0.504 mol h⁻¹ kg⁻¹ productivity, and 1003 kJ kgCO₂⁻¹ energy. The energy-productivity tradeoff is therefore milder than the purity-recovery tradeoff once purity and recovery constraints are imposed.

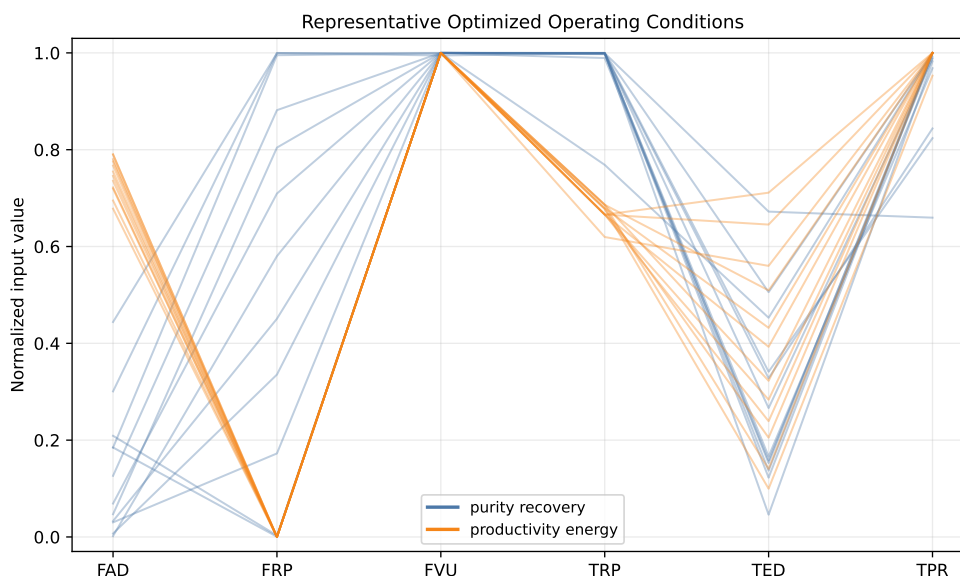


Figure 9: Representative optimized operating conditions selected from the surrogate fronts.

7 Conclusions

The project converts the successful gPROMS VPSA simulations into a clean scalar dataset, benchmarks tuned machine-learning surrogates, and uses the best model for multi-objective optimization. The RBF SVR gives the best Pareto-aware selection score and very high held-out accuracy across the four KPIs. NSGA-II reveals two operating regimes: a broad purity-recovery tradeoff with strong competition between objectives, and a narrower productivity-energy front under purity and recovery constraints. The workflow is reproducible through the scripts in `src/` and the generated CSV artifacts in `data/`.

AI Use Acknowledgement

Generative AI was used to assist with code organization, figure generation, and report drafting. The data, interpretation, and final responsibility remain with the student.

References

- [1] Zhenguang Wang, Yuanhui Shen, Donghui Zhang, Zhongli Tang, and Wenbin Li. A comparative study of multi-objective optimization with ann-based vpsa model for CO₂ capture from dry flue gas. *Journal of Environmental Chemical Engineering*, 10(4):108031, 2022.